

PAPER_173: Modular Compressed MUGE — 9-Term Mathematical Decomposition

Whitepaper §2.4-E | Thread 381a8fe7 | Session 48

Abstract

The Modular Unified Gravity Equation (MUGE) in compressed form decomposes gravitational dynamics into 9 independent sub-terms. Each term captures a distinct physical contribution: Newtonian base, cosmological expansion, magnetic suppression, environmental context, Ug contributions, cosmological constant, quantum corrections, fluid dynamics, and density perturbations.

UQFF Discovery: Novel application of UQFF calibration constants ($\kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

$$F_U(r, t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{bi}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$g_{\text{UQFF}}(r) = g_{\text{MUGE}}(r) \cdot \left(1 - [SSq] \cdot U_{bi} / F_U(r, t)\right), \quad [SSq] = 0.57$$

1. MUGESystem Struct

```
struct MUGESystem {
    std::string name;
    double I;           // Moment of inertia [kg·m²]
    double A;           // Area / cross-section [m²]
    double omega1, omega2; // Angular frequencies [rad/s]
    double Vsys;        // System volume [m³]
    double vexp;        // Expansion velocity [m/s]
    double t;           // System age [s]
    double z;           // Redshift
    double ffluid;      // Fluid frequency [Hz]
    double M;           // System mass [kg]
    double r;           // Characteristic radius [m]
    double B, Bcrit;    // Magnetic field and critical field [T]
    double rho_fluid;   // Fluid density [kg/m³]
    double g_local;     // Local gravitational acceleration [m/s²]
    double M_DM;        // Dark matter mass [kg]
    double delta_rho_rho; // Density perturbation ratio
};
```

2. Nine Sub-Terms

Term 1 — Newtonian Gravitational Baseline

compressed_base = $G \times M / r^2$

Constants: $G = 6.67430\text{e-11}$

Test: $M=1.989\text{e30 kg}$, $r=1.496\text{e11 m}$? $\sim 0.0059 \text{ m/s}^2$

Term 2 — Hubble Expansion

$\text{compressed_expansion} = 1 + H_0 \times v_{\text{exp}}$
 $H_0 = 2.269 \times 10^{-18} \text{ s}^{-1}$ ($\sim 70.1 \text{ km/s/Mpc}$)
At $t=0$: $\text{expansion} = 1.0$

Term 3 — Magnetic Suppression (Super Adjustment)

$\text{compressed_super_adj} = 1 - B/B_{\text{crit}}$
Test: $B=1 \times 10^{10} \text{ T}$, $B_{\text{crit}}=1 \times 10^{11} \text{ T}$? 0.9
Above B_{crit} : approaches 0 (magnetic quench)

Term 4 — Environmental Factor

$\text{compressed_env} = 1.0$ [placeholder for ISM/nebular environment]

Term 5 — Ug Contributions Sum

$\text{compressed_Ug_sum} = 0.0$ [Ug interface placeholder for future coupling]

Term 6 — Cosmological Constant Term

$\text{compressed_cosm} = ? \times c^2 / 3$
? = $1.1 \times 10^{-52} \text{ m}^2$ (dark energy)
 $c = 3 \times 10^8 \text{ m/s}$
? $\text{compressed_cosm} = 1.1 \times 10^{-52} \times 9 \times 10^{16} / 3 \sim 3.3 \times 10^{-37} \text{ m/s}^2$

Term 7 — Quantum Correction

$\text{compressed_quantum} = (? / ?x_p) \times ?? \times (2p / t_{\text{Hubble}})$

Parameters:

? = $1.0546 \times 10^{-34} \text{ J}\cdot\text{s}$
? x_p = $?x \times ?p = 1 \times 10^{-68} \text{ J}\cdot\text{m}$ (minimal uncertainty product)
?? = $\text{integral_psi} = 2.176 \times 10^{-18}$ (ground state energy proxy)
 t_{Hubble} = $4.35 \times 10^{17} \text{ s}$

? quantum = $(1.0546 \times 10^{-34} / 1 \times 10^{-68}) \times 2.176 \times 10^{-18} \times (2p / 4.35 \times 10^{17})$
= $1.0546 \times 10^{-34} \times 2.176 \times 10^{-18} \times 1.443 \times 10^{-17}$
 $\sim 3.312 \times 10^{-1}$

Term 8 — Fluid Dynamics

$\text{compressed_fluid} = \rho_{\text{fluid}} \times V_{\text{sys}} \times g_{\text{local}}$
Test (SGR1745): $\rho_{\text{fluid}}=1 \times 10^{-15}$, $V_{\text{sys}}=4.189 \times 10^{12}$, $g_{\text{local}}=10.0$
? $\text{compressed_fluid} = 1 \times 10^{-15} \times 4.189 \times 10^{12} \times 10 = 4.189 \times 10^{-2}$

Term 9 — Density Perturbation

$\text{compressed_perturbation} = M \times (\Delta \rho / \rho + 3 \times G \times M / r^3)$
Captures dark matter and baryonic density contrast effects.

3. Full Compressed MUGE

$\text{compressed_MUGE} = \text{base} \times \text{expansion} \times \text{super_adj} \times \text{env} \times (1 + \text{Ug_sum})$
 $+ \text{cosm} + \text{quantum} + \text{fluid} + \text{perturbation}$

Expected (SGR1745):

base $\sim G \times 2.984e30 / 1e8 \sim 1.99e11$

expansion $\sim 1 + 2.269e-18 \times 1e3 \sim 1.0$

fluid = $4.189e-2$

perturbation $\sim M \times d?$ terms

Total ? $\sim 1.782e39$ (from unit test)

4. Canonical System Test Values

System	compressed_MUGE [m/s^2]
SGR 1745-2900	$\sim 1.782e39$
Sagittarius A*	$\sim 1.782e39 \times (M_{\text{SgrA}} / M_{\text{SGR}})$
Student Guide	cosmological scale

5. Relationship to SOURCE4

The 9-term compressed MUGE directly corresponds to the `compute_compressed_MUGE_SOURCE4()` function in `MAIN_1_CoAnQi.cpp` (lines 25623-26026, namespace SOURCE4). The thread 381a8fe7 version provides the modular sub-term decomposition enabling independent validation.

6. References

- MUGE.h/cpp (thread 381a8fe7)
- UnitTests.cpp lines 1-200 (test_compute_compressed_MUGE expected=1.782e39)
- PAPER_174 (resonance MUGE, same MUGESystem struct)
- SOURCE4 integration commit 3e66d94